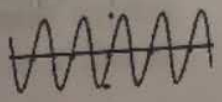
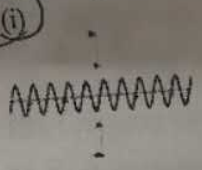
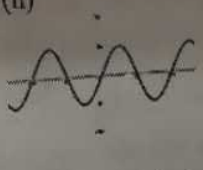
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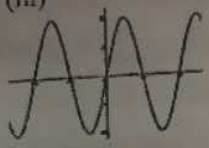
1. **(11 PTS)** Circle correct answers.

(a) *(1)* Optoelectronics engineers illuminate a Pt surface with ultraviolet light represented by the wave shown to the right, but it does not eject electrons. Which light wave below is more likely to eject electrons?



(i) 

(ii) 

(iii) 

(iv) none of these

(b) *(2)* An aqueous solution of Cl_2 is added to separate test tubes containing F^-, Br^-, or I^- ions. In which case will a halogen displacement reaction occur?

(i) Br^- and I^- only (ii) Br^- only (iii) F^- only (iv) all three cases

(c) *(2)* At a battery processing plant, you wish to separate Cu^+, Ni^{2+}, and Mn^{2+} ions based on their magnetic properties. Which set lists the ions with correct ground-state electron configurations and in order of increasing number of unpaired electrons?

(i) Cu^+ $[Ar]\,4s^1\,3d^9$ < Ni^{2+} $[Ar]\,4s^2\,3d^6$ < Mn^{2+} $[Ar]\,4s^2\,3d^3$
(ii) Ni^{2+} $[Ar]\,4s^2\,3d^6$ < Cu^+ $[Ar]\,4s^1\,3d^9$ < Mn^{2+} $[Ar]\,4s^2\,3d^3$
(iii) Cu^+ $[Ar]\,3d^{10}$ < Ni^{2+} $[Ar]\,3d^8$ < Mn^{2+} $[Ar]\,3d^5$
(iv) Mn^{2+} $[Ar]\,3d^5$ < Ni^{2+} $[Ar]\,3d^8$ < Cu^+ $[Ar]\,3d^{10}$

(d) *(2)* The nuclear chain reaction in Canada's CANDU reactors is controlled by adding natural boron, which consists of two isotopes, ^{10}B (10.013 amu) and ^{11}B (11.009 amu). You test a sample containing the $^{10}B^{3-}$ anion. What is the percent abundance of the ^{10}B isotope, and what are the numbers of subatomic particles in a $^{10}B^{3-}$ anion?

(i) 19.9%; 5 p^+, 5 n^0, 8 e^- (ii) 19.9%; 5 p^+, 5 n^0, 2 e^- (iii) 19.9%; 5 p^+, 10 n^0, 8 e^- (iv) 80.1%; 5 p^+, 5 n^0, 8 e^-

B: 10.81

$(1-x)11.009 + x\,10.013 = 10.81$

(e) *(2)* Petroleum engineers are applying "direct lithium extraction" to recover valuable lithium from oilfield wastewater by separating Li^+ ions from other ions like Na^+. Which true statement about atomic properties allows Li^+ to be differentiated from Na^+?

(i) Li^+ ion has more protons than Na^+ ion.
(ii) Li^+ ion has a smaller ionic radius than Na^+ ion.
(iii) Li atom has a lower first ionization energy than Na atom.
(iv) The outermost electron experiences a weaker effective nuclear charge in Li atom than in Na atom.

(f) *(2)* A chemical engineer is evaluating methods for producing hydrogen as an alternative fuel. Which of these reactions will not generate hydrogen gas as a product?

(i) Steam-hydrocarbon reforming process
(ii) Haber process
(iii) Reaction of sodium and water
(iv) Reaction of carbon monoxide and water

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2. **(11 PTS)** In northern Ontario, extreme seasonal temperature swings can deform steel rail lines. To join rail gaps, Canadian National Railways maintenance crews use on-site thermite welding to produce molten Fe in an exothermic reaction, with calcium fluoride added as a flux to improve fluidity and to separate the products.

(a) *(3.5)* Balance the thermite reaction using whole-number coefficients. ✓

2 Al(s) + 1 Fe_2O_3(s) → 1 Al_2O_3(s) + 2 Fe(l).

The reducing agent is Al(s) ✓, which is converted to the product Al^{3+}(s) X / Al_2O_3.

Write systematic names or provide the chemical formula for:

3

Fe_2O_3 iron (III) oxide ✓

Al_2O_3 aluminium oxide ✓

calcium fluoride CaF_2 ✓

(b) *(2.5)* Al is a moderately expensive metal, so you want to minimize materials costs. To repair a rail joint, you need 838 g of Fe(l), but the percent yield of the thermite reaction can be as low as 50%. Calculate the minimum mass (in g) of Al(s) needed to ensure that you produce this amount of Fe(l).

$838 \div 55.85 = 15\ mol$ ✓

$15 \times 100 \div 50 = 30\ mol$ ✓

2.5

$Al_{(s)} : Fe_{(l)} = 1:1$ ✓ $30 \times 26.98 = 809.4\ g$ ✓

so 30 mol $Al_{(s)}$ needed

(c) *(2)* Successive ionization energies for Al are listed:

IE_1	IE_2	IE_3	IE_4
578 kJ/mol	1817 kJ/mol	2745 kJ/mol	11577 kJ/mol

Write an equation corresponding to IE_4 and explain why it is so much larger than IE_3.

more energy required to ✓ over through it.

$Al^{3+}_{(g)} = Al^{4+}_{(g)} + e^-$ ✓

2

there is 3 e^- on the outer shell of Al, the IE_4 jumped to next shell which is closer to the nucleus with stronger attraction so

(d) *(3)* The melting points of ionic solids can be related to magnitudes of lattice energies. Rank these substances in order of increasing mp: Al_2O_3, calcium fluoride, calcium chloride.

(lowest mp) Calcium chloride ($CaCl_2$) < Calcium fluoride (CaF_2) < Al_2O_3 (highest mp) $CaCl_2$

2

Justify your ranking.

Calcium chloride and calcium fluoride have same number of bonds and Al_2O_3 have more bonds than both of them so ~~it's stronger~~ it has highest mp X

~~chlorine (of chloride ion) have larger atomic radius which means bonds are longer so it's easier to break~~ ? ~~than fluorine ion~~

~~Calcium chloride have~~ Cl^- have larger radius so less attraction, so $CaCl_2$ have lower mp ✓

9.5

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3. **(11 PTS)** Aerospace engineers working with the Canadian Space Agency are designing space telescopes to determine compositions of a hot massive stars by detecting their emission spectra.

(a) *(3)* You observe a green emission line at 541.2 nm which you assign to an electronic transition in a H atom from some unknown $n_{initial}$ to $n_{final} = 2$. Determine $n_{initial}$ (show work) and explain why this result invalidates your hypothesis that the line comes from a H atom.

$E = \frac{hc}{\lambda} = \frac{6.626\times10^{-34} \times 2.998\times10^{8}}{541.2\times10^{-9}} = 3.671\times10^{-19}\ J$ ✓

$E = -R_H Z^2\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$

$-3.671\times10^{-19} = -2.179\times10^{-18} \times 1^2\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$

$\frac{1}{4} - \frac{1}{n_i^2} = \frac{-2.179\times10^{-18}}{-3.671\times10^{-19}}$

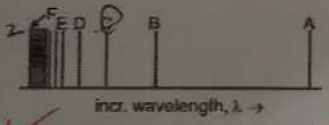
$\frac{1}{4} - \frac{1}{n_i^2} = 5.938$

$\frac{1}{n_i^2} = 0.647$

$n_i = 1$ calc error?

1.5

(b) *(2)* In fact, the line at 541.2 nm arises from a different one-electron species with $Z = 2$. It corresponds to line C in a series of emission lines due to transitions from various initial states to a common final state corresponding to the *third excited state*.



The one-electron species is: He^+ H_0. For line C, $n_{initial} =$ 7 ✓ and $n_{final} =$ 2 + 4.

1

Line A (at 1012.4 nm) appears in this region of the electromagnetic spectrum (circle one):
(infrared) ✓ microwave ultraviolet visible

(c) *(6)* The strongest line (at 468.6 nm) detected is due to a transition from a 4f to a 3d orbital. Specify all possible quantum numbers consistent with a 4f orbital:

$n =$ 4 ✓

$\ell =$ 3, 2, 1, 0 ✗ $\ell = 3$

$m_\ell =$ −3, −2, −1, 0, +1, +2, +3 ✓

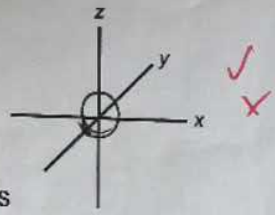
$m_s =$ $-\frac{1}{2}, \frac{1}{2}$ ✓

1.5

Specify the number of nodes consistent with a 3d orbital:

no. of radial nodes = 0 5 ✗ no. of angular nodes = 5 + 2

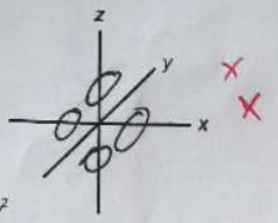
Draw the boundary surfaces of the specified orbitals and indicate relative phases (signs) of the wavefunctions in different regions of space.



3s

✓ ✗

0.5



$3d_{x^2-y^2}$

✗ ✗

Explain if the 3s and $3d_{x^2-y^2}$ orbitals have the same or different energy in a H atom.

they have different energy since the position is different.

– same E, one-e^- species

4.5

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